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# Developing a Highly Secure and High Capacity LSB Steganography Technique using PRNG

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***Abstract*— Being a modern way of secret messaging, steganography has become one of the trending topic for researchers and scientists. Using image in data secrecy is most common among all other ways of digital communication. In steganography, a carrier object behaves as a cover for the secret object such that intruders are not aware of its existence. Thus, it is important to create a steganography technique which is highly secure and also has a high data hiding capacity. In the technique proposed, each pixel has an enhanced capacity of 3-6 bits. This capacity is directed by the most significant bits of the indicator channel. Use of PRNG has made this technique more secure. The results achieved by the experiment signify the profoundness of this technique.**

***Keywords—steganography, stego-image, LSB, PIT, PRNG, statistical steganalysis.***

## I. Introduction

Steganography, literally translated to 'covered writing' is an age-old practice of hiding useful information in plain sight. It has been existing since ancient times when some secret messages were traded between people. Some people would tattoo the message on a bald head and then send the person when his/her hair grew back, or using microdots and invisible ink to convey messages [1]. Today however, digital media is used to hide the secret encoded messages. A number of digital media are present on web pages across the internet. Steganography provides a means to conceal sensitive information within images, videos and also audio files. Cryptography focuses on transforming a message so that it cannot be understood by people it wasn't meant for, while steganography is meant to shroud the existence of the information or secret data [2]-[3]. People all over the world have been communicating extensively through internet and the need to transfer sensitive information without it dropping into wrong hands has increased tremendously. Plenty of steganography algorithms and techniques have been suggested and developed for the transfer of information. Based on the carrier file, steganography can be classified into the following:

- Image steganography: The carrier object is an image and the secret object is hidden in the bits of cover object pixels.
- Video steganography: A video is fundamentally a collection of images. Data is thus hidden inside these images but the cover file is a video.
- Network steganography: Communication protocols are used as cover media. Typically, all the protocols have a header that is used to convey certain information. Most of the times there are some unused bits present in the headers which are used to store secret message data [4].
- Audio steganography: The hiding medium is an audio file; this method has gained popularity since the use of Voice over IP (VoIP) has increased.

Text steganography: This technique hides the text message behind another text file. White spaces, or mis-spelled words during a certain time duration are used to convey hidden messages.

## II. Image Steganography

Digital images are widely used today for steganographic purposes thanks to the presence of a variety of images across web pages found on the internet. A traditional steganography system with no cryptography is shown in Fig. 1 [11]. Eight bits are used to depict the color intensity in grayscale or monochrome pictures whereas the true color or RGB model uses 24 bits for each pixel. The basic essence of this steganography lies in the fact that the resolution of human eye is not enough to detect small or minute changes in the pattern or intensity of colors. Therefore, text or graphic messages can be easily hidden within image carrier files without garnering unwanted attention [5].

Image steganography techniques have been divided into two main domains and different methods have been developed for each domain.

Spatial domain: All methods available directly manipulate some bits in the pixel values. These methods have the advantage of being simpler and have a higher data hiding capacity but are vulnerable to simple attacks and are less robust [6]. Some prevalent methods in this section are:

1. Edge based embedding methods

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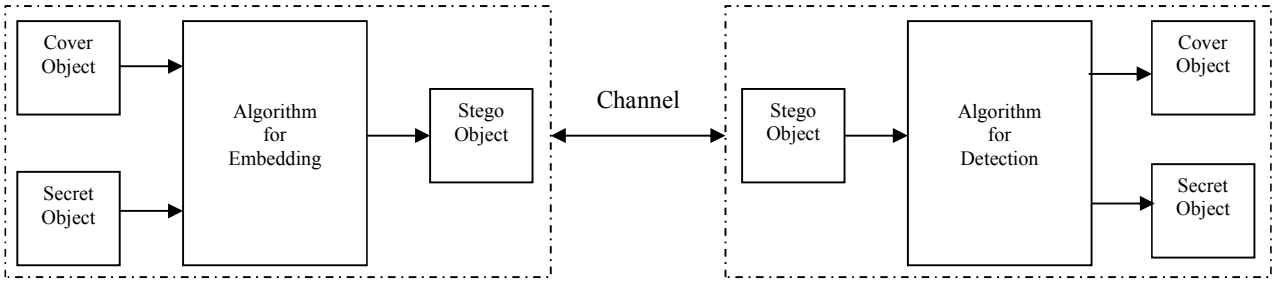


Fig. 1. Illustration of a steganography system without cryptograph

2. Least Significant Bit (LSB)
3. Pixel Indicator Technique (PIT)
4. Pixel Value Differencing (PVD)

Transform Domain: This includes methods that are more complex but information is concealed in parts of the carrier that are less vulnerable to cropping or compression [6]. This makes the transform domain methods more robust and secure as compared to spatial domain. Several techniques have been developed in transform domain for steganography which are discussed in [7]. These techniques can be classified into various methods such as:

1. Discrete Cosine Transformation
2. Discrete wavelet Transformation
3. Discrete Fourier Transformation

Steganography algorithms are judged on the basis of many measures or parameters. These parameters include capacity, robustness, perceptual transparency (security), computational complexity and many others. Methods for image steganography are categorized as per the parameters discussed in [14] and [15]. Capacity and security are the two most used parameters for analyzing the steganographic techniques and methods. Mean Square Error (MSE) is usually calculated for the image before and after encoding the message. This calculated value is used for computation of Peak Signal to Noise Ratio (PSNR). The mathematical expression of the two are given below.

$$MSE = \frac{1}{R*C}\sum_{i=1}^{R}\sum_{j=1}^{C}(x_{ij}-x'_{ij})^2 \quad (1)$$

In (1), R stipulates the rows in the image; C stipulates the columns in the image; $x_{ij}$ indicates the carrier image and $x'_{ij}$ indicates the stego-image.

$$PSNR = 10\log_{10}\left[\frac{I^2}{MSE}\right](dB) \quad (2)$$

In (2), I indicates the maximum possible value of a pixel present in the image.

PSNR is the ratio that is used for the indication of quality of the images. Higher PSNR favors higher quality.

Steganography is used for a variety of applications which include but are not limited to secret communication across the web, fingerprinting and digital watermarking which is used for ownership purposes. Copywrite protection can also employ steganography methods to achieve its goal.

## III. Related Work

The most basic and simplest image steganography technique is LSB technique wherein the least significant bits in the pixel values are replaced by the secret message bits. Because of the ease and simplicity of LSB many attempts have been made to enhance the technique for increasing the security of the data by combining cryptology. A 5.8% increase in the PSNR value was achieved by Zhou, Gong, Fu and Jin when traditional LSB was combined with Rivest-Shamir-Adlemam (RSA) technique for cryptology [8]. Another algorithm proposed to adopt distinguishing directions for embedding the secret bits [9]. Security was also improved by combining LSB and DCT as was shown by Kaur and Kochhar [10].

Another popular technique is the Pixel Indicator Technique (PIT) where the for each pixel a channel (RGB) is specified known as the indicator channel. The LSB of this channel signify number of message bits are concealed in the other two channels. Adnan Gutub compared PIT with Stego Color Cycle (SCC) on the basis of number of hidden bits and proved that PIT is better [11]. SCC was introduced to strengthen security of traditional LSB. A very clever improvisation for hiding the data in the pixels was suggested that is meant to confuse the steganalysis attempts thereby, promoting secure exchange of secret data [13].

A novel approach was introduced by Kukapalli, Rao and Reddy to utilize the MSB bits. The indicator channel's MSB bits were utilized to signify number of hidden bits in the channels. Also, the security was tightened by using Blowfish algorithm [12]. Random bit substitution by using a PRNG and performing XOR to increase the security was proposed by Ali, Sohrawordi and Uddin. The results showed no improvement in PSNR values but the method boosted the security because of the randomness [5].

Another method was proposed that utilizes the MSB for mapping [16]. Drawback of this method was that the secret data should only contain alphabets.

## IV. Proposed Method

The basic LSB technique though the simplest, is prone to many attacks and the presence of embedded data can easily be detected by observing the histogram of the images. The hidden information can also be detected and extracted by using many statistical steganalysis techniques [17]-[18]. The proposed method does replace the LSB bits of the cover image but the algorithm makes it difficult to detect the actual message data, enhancing the security of the system. The indicator channel is


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chosen randomly by the PRNG and the number of bits to be stored depends on the MSB bits that are already available in the image. The MSB technique discussed in [12] hides 1 bit of data for '1' and no data for the occurrence of '0'. The proposed method aims to increase the data capacity by embedding 2 data-bits for the occurrence of '1' and 1 data-bit for occurrence of a '0'. Traditional LSB has a lower computational complexity than the proposed technique.

The proposed method is illustrated below in Figure 2.

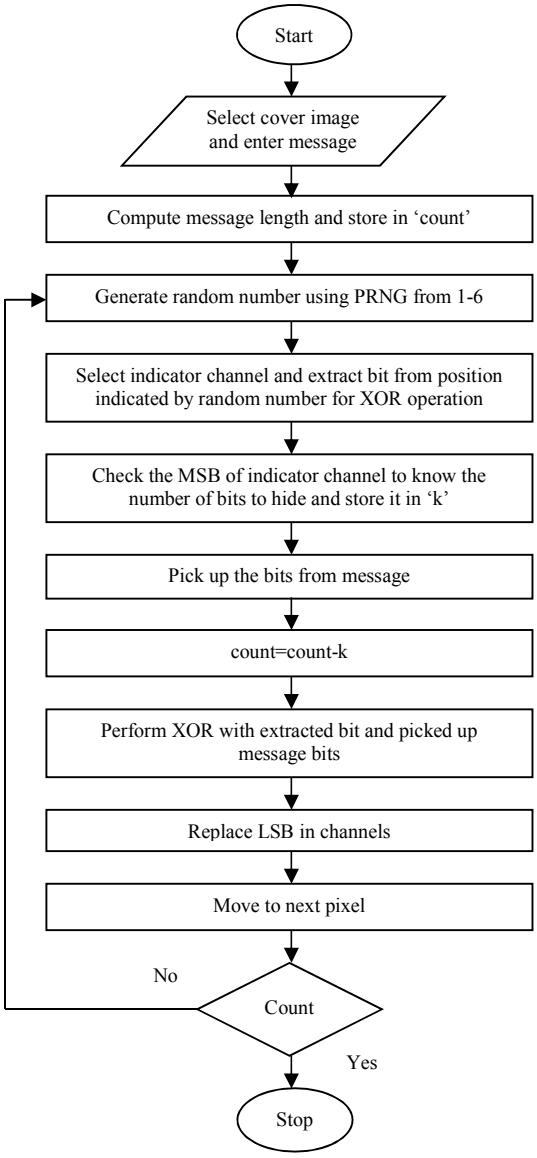


Fig. 2. Flow diagram of proposed method

Algorithm for hiding the data:

1. Select the carrier image and ask the user for message data, which is converted to binary.
2. Generate a number using Pseudo Random Number Generator (PRNG).
3. Based on the number generated by PRNG, the indicator channel is selected from the pixel accordingly as shown in table I. Also, the number generated by PRNG indicates the bit position in the indicator channel for the random bit to be selected for performing the XOR operation with the message bits.

TABLE I. SELECTION OF INDICATOR CHANNEL

| Number generated by PRNG | Indicator channel |
|---|---|
| 1 or 4 | Red |
| 2 or 5 | Green |
| 3 or 6 | Blue |

4. The 3-bit MSB of indicator channel indicates the number of pixels to be hidden in each channel as shown in table II.

TABLE II. NUMBER OF HIDDEN BITS IN EACH CHANNEL

| Indicator channel MSB bits | Hidden bits in Red channel | Hidden bits in Green channel | Hidden bits in Blue channel |
|---|---|---|---|
| 000 | 1 | 1 | 1 |
| 001 | 1 | 1 | 2 |
| 010 | 1 | 2 | 1 |
| 011 | 1 | 2 | 2 |
| 100 | 2 | 1 | 1 |
| 101 | 2 | 1 | 2 |
| 110 | 2 | 2 | 1 |
| 111 | 2 | 2 | 2 |

5. The XOR of message bits and bit extracted from PRNG is done.
6. The resultant is hidden in the LSB of channels as indicated by the MSB of indicator channel.
7. The steps are reiterated for the subsequent pixels till the value of count is not zero.

For example, the message bits are 0100010110 and the randomly generated numbers are 2, 6, 3 and so on. The embedding process is illustrated below for two pixels in table III. Under the column 'Pixel values', the three bold digits represents the MSB of indicator channel and the underlined digit is the bit used for XOR operation.

## V. EXPERIMENTAL RESULTS AND ANALYSIS

PyCharm was used to design and implement the proposed technique by using Python. The widely available 24-bit images have been chosen as the cover images for the calculation and analysis of the results. The cover images are illustrated in Fig. 3. Size of the images is 512x512 and the secret message used is "HelloWorld".

| Lena | Peppers | Baboon |
|---|---|---|

Fig. 3. Cover images

Another program was written for computing the values of MSE and PSNR of the images. These values are compared with values of existing techniques. The stego images with their respective values of MSE and PSNR are depicted in Figure 4.


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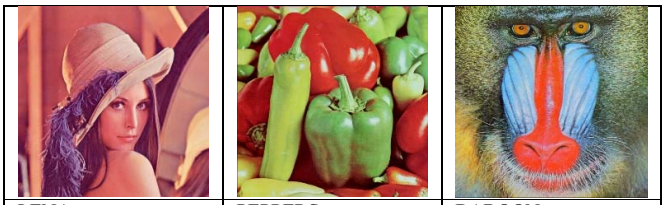
TABLE III. PROCESS OF EMBEDDING THE MESSAGE IN THE PIXELS

| Pixel Values | | Random number and bit position | Indicator channel and its MSB | Number of bits to be hidden in each channel | XOR bit | Ex-or with message | Substituting at LSB bits |
|---|---|---|---|---|---|---|---|
| Pixel 1 | R-01000110 | 2<br>(010) | Green<br>100 | 4 bits | 0 | 0⊕0=0<br>1⊕0=1 | 01000101 |
| | G-10010011 | | | | | 0⊕0=0 | 10010010 |
| | B-11001100 | | | | | 0⊕0=0 | 11001100 |
| Pixel 2 | R-11100010 | 6<br>(110) | Blue<br>101 | 5 bits | 0 | 0⊕0=0<br>1⊕0=1 | 11100001 |
| | G-00101010 | | | | | 0⊕0=0 | 00101010 |
| | B-10101010 | | | | | 1⊕0=1<br>1⊕0=1 | 10101011 |



| LENA | PEPPERS | BABOON |
|---|---|---|
| MSE-0.0001055<br>PSNR-87.896 | MSE-0.0000851<br>PSNR-88.826 | MSE-0.0000814<br>PSNR-89.025 |

Fig. 4. Stego images and their values of MSE and PSNR

Security and capacity of the proposed method are juxtaposed with that of LSB technique as well as with that of random bit substitution method proposed in [5]. The values of MSE and PSNR of all the aforementioned methods are given in table IV and V respectively. All the stego images have a reasonably high value of PSNR indicating high quality.

TABLE IV. COMPARISON OF MSE

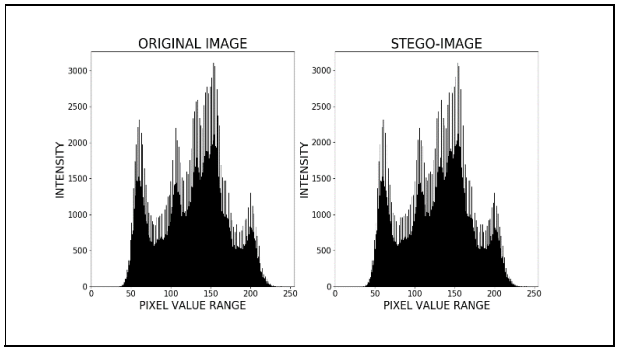
| | 1-bit LSB | 2-bit LSB | Random bit substitution | Proposed method |
|---|---|---|---|---|
| Lena | 0.0000559 | 0.0001436 | 0.00004196 | 0.0001055 |
| Peppers | 0.0000419 | 0.0001462 | 0.00002670 | 0.0000851 |
| Baboon | 0.0000534 | 0.0001220 | 0.02022934 | 0.0000814 |

TABLE V. COMPARISON OF PSNR

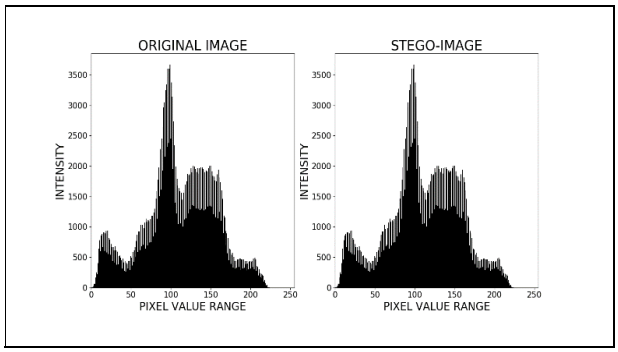
| | 1-bit LSB | 2-bit LSB | Random bit substitution | Proposed method |
|---|---|---|---|---|
| Lena | 90.65 | 86.56 | 90.18 | 87.896 |
| Peppers | 91.90 | 86.48 | 90.09 | 88.826 |
| Baboon | 90.85 | 87.26 | 61.48 | 89.025 |

The traditional LSB substitution is the least secure method because it lucidly changes the LSB with the secret message bit. Random bit substitution claims to be more secure than LSB due to the random bit that is used for X-OR. The proposed method is the most secure given the randomness of X-OR bit as well as that of the indicator channel. Also, unlike the traditional LSB and random bit substitution the capacity of the channels is not fixed. The capacity of any pixel is dictated by the MSB of the indicator channel's most significant bit, which is chosen on the basis of the value generated by PRNG. This computational complexity boosts security and makes it difficult to extract the message easily.

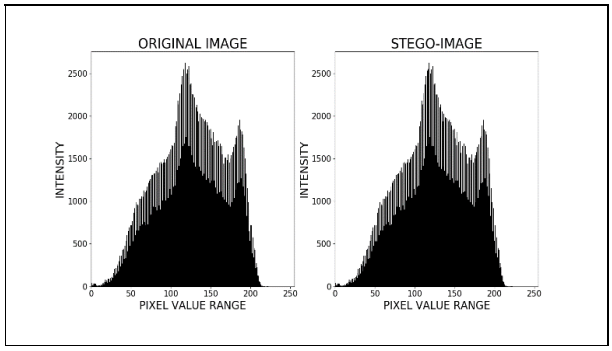
Histograms are most commonly used for analyzing the security of the technique. Most steganalysis methods rely on histograms for detecting hidden information. Therefore, it is crucial that there should be no obvious or blatant changes in the histogram of the stego-image when compared to the cover image. The histograms of carrier image as well as stego-image are illustrated in figure 5.




(i)




(ii)




(iii)

Fig. 5. Histogram of cover and stego-images: (i) Lena, (ii) Peppers, (iii)Baboon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Capacity of any technique increases as the quantity of information bits stored in any pixel increases. It is a crucial decision since it affects the transparency of the image. The hidden bits should not bring about any blatant changes in the image otherwise steganography loses its advantage of being stealthy and covert. The proposed method hides a minimum of 3 bits per pixel and a maximum of 6 bits per pixel depending upon the MSB of the indicator channel. The technique proposed has a higher capacity than that of 1-bit LSB as well as random bit substitution method; both of the above storing 3 bits per pixel. The lower PSNR of 2-bit LSB substitution is compensated by its capacity; as it is capable of hiding 6 bits per pixel.

Hence, the proposed method provides a higher capacity as well as enhanced security owing it to the elaborate algorithm. Furthermore, the security can also be surged by using an encryption technique.

## VI. Conclusion And Future Work

For steganography methods, security and capacity are the two parameters which are often in competition. The proposed method avails heightened security and increased capacity too. Weakness of the traditional LSB technique is the repetitive pattern in which it stores the data. The proposed method does not follow a repetitive pattern and takes advantage of the randomness of PRNG to securely hide the information. Unlike the traditional LSB technique the original message is not hidden in the carrier image. Undoubtedly, the proposed technique is superior in capacity to the 1-bit LSB substitution and random bit substitution. The capacity of 2-bit LSB is comparable to the proposed method but the mentioned LSB system lacks security as indicated by PSNR values. Hence, the proposed technique proves to be the best among the discussed methods for a secure communication over the internet. As an example of future work, we plan to add encryption techniques for further tightening the security.

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